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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION October 2026
Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEPHY303 - LASER, NON-LINEAR OPTICS AND FIBER OPTICS
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A: Short Answer Questions
Answer any 10 questions. Each question carries 2 marks

1 State **any two** key operating characteristics of a ruby laser. [U]/[CO1]

Pulsed Laser Output/Spiking/ Three level laser. **(2 Marks)**

2 Explain Stimulated emission process in a laser system. [U]/ [CO1]

An incident photon with energy matching the transition interval ($h\nu = E_2 - E_1$) interacts with an atomic system in an excited state, thereby inducing a radiative transition to a lower energy level. **(2 Marks)**

3 State how acceptor doping alters the position of the Fermi level in a semiconductor. [U]/[CO1]

Acceptor doping introduces acceptor energy levels just above the valence band, creating an excess of holes (majority carriers) this shifts the Fermi level downward closer to the valence band, away from the middle of the forbidden energy gap. **(2 Marks)**

4 Under what condition does resonance occur in an open resonator? [U] / [CO1]

Resonance in an open resonator occurs when an integral number of half-wavelengths covers the length between the reflecting mirrors.

OR

$L = \frac{q\lambda}{2}$, where q is an integer and λ is the wavelength of light. **(2 Marks)**

5 Explain the working principle of a mechanical Q-switch in a laser cavity. [U] / [CO2]

One of the end mirrors of the laser resonator is kept rotating rapidly about an axis perpendicular to the axis of the resonator. The mirror is only instantaneously aligned parallel to the output reflector. **(1 Mark)**

The flash lamp is timed to fire so that the mirror aligns only when the population inversion reaches its maximum. **(1 Mark)**

6 What is a laser amplifier? [K]/ [CO2]

A laser amplifier is a device used to obtain higher-intensity laser beams by directing a laser pulse through a gain medium (such as a laser rod) that does not have external reflectors. **(1 Mark)**

The amplifying rods used have uncoated ends or are treated with antireflecting coatings and the amplifier is pumped with its own flash lamp. **(1 Mark)**

7 State why second-harmonic generation (SHG) cannot occur in centrosymmetric media. [U] / [CO3]

In a centrosymmetric medium, spatial inversion transforms the electric field and second-order polarization such that $E \rightarrow -E$ and $P^{(2)} \rightarrow -P^{(2)}$. **(1 Mark)**

Substituting this into the nonlinear relation gives $-P^{(2)} = P^{(2)}$, which forces $P^{(2)} = 0$ and consequently $\chi^{(2)} = 0$, prohibiting second-harmonic generation. **(1 Mark)**

8 Write down the frequency relation governing the generation of an idler wave in parametric processes and define each term. [U]/ [CO3]

$\omega_i = \omega_p - \omega_s$ **(1 Mark)**

ω_p -Pump Frequency, ω_s -Signal Frequency, ω_i -Idler Frequency **(1 Mark)**

9 What is the purpose of an erbium-doped fibre amplifier (EDFA) in long-distance optical fibre communication? [U]/[CO4]

The light pulse attenuates as it travels. Rather than using optoelectronic regenerators, modern networks amplify the optical signal directly using inline Erbium-Doped Fiber Amplifiers (EDFAs), which provide ≈ 20 dB of optical gain within the 1530 - 1570 nm band. **(2 Marks)**

10 How is photosensitivity at ~240nm utilized to fabricate periodic gratings in germanium-doped silica fibres? [U]/ [CO4]

Periodic variation of refractive index along the dispersive fibre element can be induced in germanium doped silica by forming an interference pattern within the fibre core from two coherent UV laser light beams incident at an angle to the fibre axis. **(1 Mark)**

The glass core of the fibre is photosensitive at ~ 240 nm and this wavelength may be obtained by harmonic generation from a longer wavelength laser, or it may be directly produced by a pulsed ultraviolet excimer laser. **(1 Mark)**

11 Define the acceptance angle of an optical fibre. What is meant by the acceptance cone? [U]/[CO4]

The acceptance angle is defined as the maximum angle to the fibre axis at which light can enter the core of an optical fibre from an external medium guided along the fibre via total internal reflection.

OR $n_0 \sin\theta_a = n_1\sqrt{1 - \left(\frac{n_2}{n_1}\right)^2}$ **(1 Mark)**

The acceptance cone is the three-dimensional conical volume formed by acceptance angle about the central axis of the optical fibre core.. **(1 Mark)**

12 What is meant by air-guiding type photonic crystal fibre? [K]/ [CO4]

In an air-guiding type photonic crystal fibre (also known as a hollow-core photonic bandgap fibre), the central core is an **air hole** rather than a solid silica core. **(1 Mark)**

It operates on physical principle: Photonic Bandgap Guiding (or 2D Bragg Reflection). (**1 mark)**

[2×10=20]

Part B: Short Essay Questions
Answer any 5 questions. Each question carries 6 marks

13 For a plane electromagnetic wave propagating in a rectangular cavity, show that the wave vector k is perpendicular to the electric field E(t). [U]/[CO1]

Wave Equation p (1 Mark)

$$\nabla^2 \mathbf{E} = \frac{1}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Plane wave solution to the wave equation (1 Mark)

$$E(\mathbf{r}, t) = E_0 \exp\{i(\mathbf{k}\cdot\mathbf{r} - \omega t)\}$$

Spatial standing wave components inside the cavity (2 Mark)

$$E_x(t) = E_{0x}(t)\cos k_x x \sin k_y y \sin k_z z$$

$$E_y(t) = E_{0y}(t)\sin k_x x \cos k_y y \sin k_z z$$

$$E_z(t) = E_{0z}(t)\sin k_x x \sin k_y y \cos k_z z$$

Maxwell's equation (1 Mark)

$$\nabla \cdot \mathbf{E} = 0$$

$$k_x E_{0x}(t) + k_y E_{0y}(t) + k_z E_{0z}(t) = 0$$

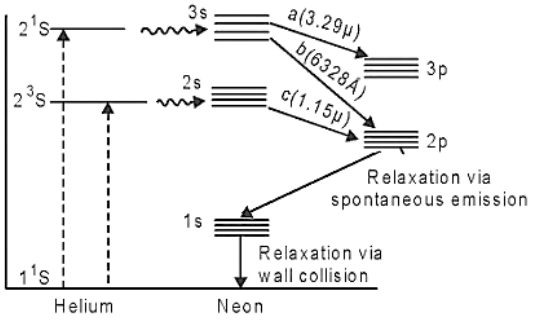
Dot Product Form and Final Conclusion (1 Mark)

$$\mathbf{k}\cdot\mathbf{E}(t) = 0$$

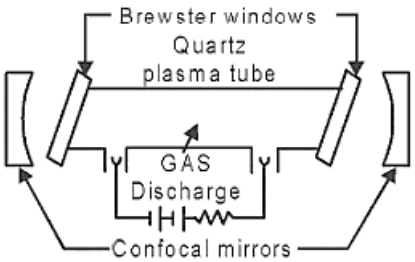
Hence the wave vector is perpendicular to the electric field vector ($\mathbf{k} \perp \mathbf{E}$)

14 Describe the construction, working principle, and key operational characteristics of a He-Ne laser with a neat energy-level diagram. [U]/ [CO1]

Energy Level Diagram **(2 Marks)**

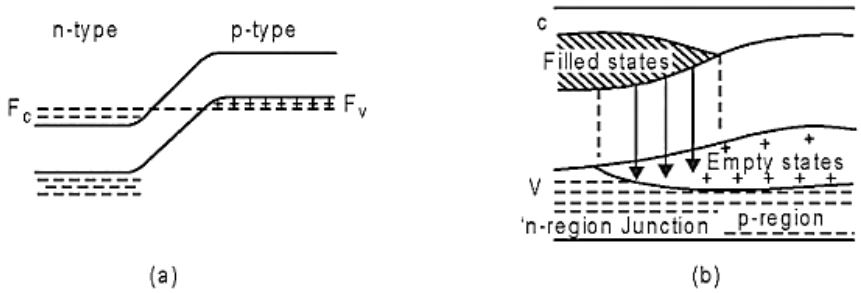


Schematic Diagram (1.5 Marks) + Explanation (2.5 Marks)



15 Explain the working of a semiconductor laser with a suitable energy level diagram. [U]/[CO1]

Energy Level Diagram (2 Marks)



Basic Principle & Requirements (3 Mark)

The semiconductor material must have a high transition probability for radiative transitions across the bandgap, exceeding non-radiative lattice transfer rates. Population inversion is achieved via carrier injection through a forward biasing of the semiconductor diode.

The p-n junction diode must be heavily doped. In the n-type region, the Fermi level F_c lies within the conduction band, while in the p-type region, the Fermi level F_v lies within the valence band.

When forward-biased, electrons are injected into the p-side and holes into the n-side, creating a high concentration of both carriers in the narrow depletion/active region. This high concentration results in a population inversion between the conduction band, and the valence band states near the junction. Electron-hole recombination in this narrow planar region releases coherent optical radiation when the injected current exceeds a threshold value.

Optical Confinement & Resonator Action (1 Marks)

Laser action is confined to the very thin junction region.

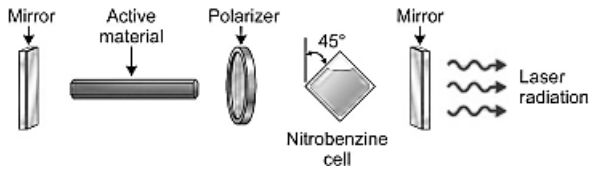
The crystal end faces are cleaved parallel to form an optical resonator, providing the necessary feedback for amplification and stimulated emission.

16 Discuss the principle and operational mechanism of a Kerr-cell Q-switch in a laser system [U]/ [CO2]

Explanation of Kerr effect (1.5 Mark)

A Kerr cell consists of an isotropic liquid (such as nitrobenzene) containing asymmetric molecules placed between parallel plates. When an external electric field is applied, the molecules tend to align parallel to the field direction and the liquid become optically anisotropic and birefringent, with refractive indices n_o and n_e proportional to the square of the applied voltage.

The Kerr Cell Shutter System-Schematic OR Explanation (1.5 Marks)



Operational Sequence (2 Marks)

The Kerr cell is activated by applying a high bias voltage (10--20 kV).

The flash lamp is fired to pump the active laser medium.

After population inversion peaks (approx. 0.5 ms), the Kerr cell voltage is switched off as suddenly as possible.

Performance Characteristics (1 Mark)

Operating as a fast electro-optic shutter, switching times in the nanosecond range. Effectively increases peak output power (e.g., from kilowatts to megawatts/gigawatts) by compressing the energy into a giant optical pulse.

17 Explain the phase-matching condition in second-harmonic generation and show mathematically how it leads to the equality of refractive indices at both frequencies ($\eta_\omega = \eta_{2\omega}$). [U]/ [CO3]

Principle of SHG and Condition for Maximum Intensity (2 Marks)

In SHG, energy transfer from the fundamental wave (ω) to the second-harmonic wave (2ω) requires constructive interference over the propagation path.

Wave number matching condition for maximum intensity:

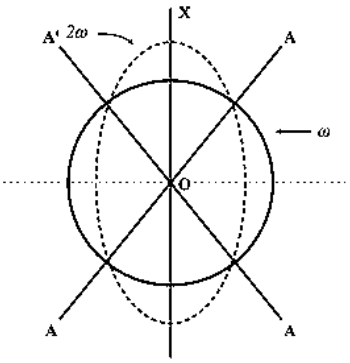
$$\frac{2k_1 - k_2}{2} = 0 \Longrightarrow k_2 = 2k_1$$

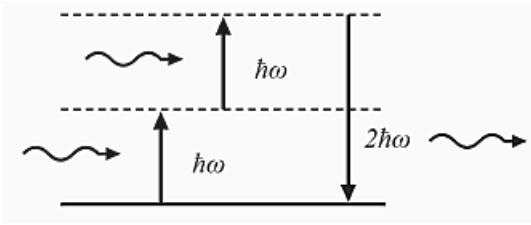
Substitution of k_1 and k_2 and arriving at the final equation (1.5 Marks).

$$\frac{2\omega\eta_{2\omega}}{c} = 2\left(\frac{\omega\eta_\omega}{c}\right)$$

$$\eta_{2\omega} = \eta_\omega$$

Refractive index ellipsoid OR Energy level representation (1.5 mark)





Physical Significance & Use of Birefringence (1 Mark)

Phase matching is achieved in anisotropic crystals by using polarization-dependent refractive indices, where ordinary ray index at frequency ω matches the extraordinary ray index at 2ω along specific propagation directions.

18 Explain the physical mechanism of self-focusing in a nonlinear medium using the Gaussian intensity profile of a laser beam. [U]/ [CO3]

Definition of self-focusing (1 Mark)

It is a non-linear process where an intense light beam modifies the refractive index of a medium such that causes the beam to focus itself, without changing its frequency.

Derivation of the expression for intensity-dependent index (2 Marks)

$$\eta = \eta_l + \eta_{nl} E_0^2$$

Explanation of Gaussin intensity profile of laser (1 Mark)

A real laser beam features a Gaussian intensity profile that peaks along the central propagation axis and drops off radially.

Expression for Self-focusing distance and Threshold Intensity (2 marks)

$$L_0 = \frac{D}{\sqrt{\eta_{nl} E_0^2}} \quad , \qquad I_{\text{thresh}} = \frac{\lambda^2}{\eta_l^2 \eta_{nl} D^2}$$

19 A step-index optical fibre has core radius a = 25 μm, n_1 = 1.520, n_2 = 1.505, and λ_0 = 2 μm. Calculate the V-number and determine whether the fibre is single-mode or multimode. Estimate the number of propagating modes. [A]/[CO4]

Given: Core radius, $a = 25\ \mu\text{m} = 25 \times 10^{-6}$ m, Core refractive index, $n_1 = 1.520$, Cladding refractive index, $n_2 = 1.505$, Free-space wavelength, $\lambda_0 = 2\ \mu\text{m} = 2 \times 10^{-6}$ m

V-number substitution and answer:

$$V = \frac{2\pi a}{\lambda_0}\sqrt{n_1^2 - n_2^2} \qquad \textbf{(1 Mark)}$$

$$V = \frac{2\pi \times (25 \times 10^{-6}\ \text{m})}{2 \times 10^{-6}\ \text{m}}\sqrt{(1.520)^2 - (1.505)^2} = \mathbf{16.7} \qquad \textbf{(1 Mark)}$$

Condition for single mode / multimode and conclusion:

$V \leq 2.405$ (Single-mode), $V > 2.405$ (Multimode) **(1 Mark)**

Since $V = 16.73$, The optical fibre is **multimode**. **(1 Mark)**

Number of propagating modes substitution and answer:

$$N \approx \frac{V^2}{2} \qquad \textbf{(1 Mark)}$$

$$N = \frac{(16.73)^2}{2} = 139.95 = \textbf{140 modes} \qquad \textbf{(1 Mark)}$$

20 Analyse how the boundary conditions of a slab waveguide restrict the possible electromagnetic field patterns and lead to the formation of discrete modes. [An]/[CO4]

Introduction to Slab Waveguide & Confinement Mechanism (1 Mark)

Confinement of light in a dielectric slab waveguide (core refractive index n_1 in between two cladding layers of lower refractive index n_2) is achieved via total internal reflection (TIR) at the boundaries. Unlike metallic waveguides, the wave amplitude does not drop to zero at the boundary; rather, an evanescent field penetrates the cladding.

Phase Shift Due to Propagation (2 Marks)

Using the core wavelength $\lambda_1 = \lambda_0/n_1$, the phase shift due to this geometrical path difference is expressed as:

$$\frac{2\pi}{\lambda_1}(2b \sin\theta)$$

Total Phase Condition (1.5 Marks)

The total phase change must be an integral multiple of 2π ($2N\pi$):

$$\frac{2\pi}{\lambda_1}(2b\sin\theta) + 2\phi(\theta) = 2N\pi$$

Formation of Discrete Modes (1.5 Mark)

Simplifying the phase condition yields the equation for discrete propagation angles:

$$2b\sin\theta + \lambda_1\frac{2\phi(\theta)}{2\pi} = N\lambda_1$$

Heree N is an integer (the mode number) and the phase shift $\phi(\theta)$ is a function of angle, only specific, discrete ray directions (θ) satisfy this condition. This restriction restricts the electromagnetic field patterns to a set of discrete guided modes.

[6×5=30]